BiasAperture — Monthly Progress & Weekly Reports Index

Period: August 2026 (2026-08-01 through 2026-08-31)
Project: BiasAperture — A Diagnostic and Evaluative Framework for Auditing Demographic Bias in Facial Analysis Systems
Program: Fusemachines AI Fellowship (AIF) 2026 (Kathmandu, Nepal)
Authors: Aaradhya Dev Tamrakar (`@AaradhyaDT`) & Tisha Manandhar (`@tiixsha`)
Supervisor: Shreejan Kisee (Teaching Assistant, Fusemachines AI Fellowship)
Current Milestone State: M1–M4 Core Engine Verified · 55/55 Tests Passing · M5 Integration in Progress

---

1. Executive Summary

During August 2026, BiasAperture advanced from an initial fellowship proposal to an audit-grade diagnostic software platform. Development was structured across four weekly sprints, covering proposal scaffolding, schema locking (M1), requirements and architecture formalization, a 20-track parallel research sprint, dataset exploration, fairness engine implementation (Fairlearn and native AIF360 backends), bootstrap confidence intervals, Chi-squared significance testing, additive Shapley surrogate feature attribution, offline HTML reporting, and a full CLI orchestrator with 55 passing unit tests.

```
August 2026 Sprint Progression:

┌──────────────────────────┐      ┌──────────────────────────┐
┌──────────────────────────┐      ┌──────────────────────────┐
│ Week 1 (Aug 01-07)       │ ──> │ Week 2 (Aug 08-14)       │ ──> │ Week 3 (Aug 15-21)
│ ──> │ Week 4 (Aug 22-31)       │
│ Inception & LaTeX Cls    │      │ M1 Schema Lock & Papers │      │ FR-001-005 & TA Stds
│      │ 20-Tracks, Math & CLI    │
└──────────────────────────┘      └──────────────────────────┘
└──────────────────────────┘      └──────────────────────────┘
```

---

2. Weekly Reports Breakdown

The complete weekly development logs are divided into dedicated reports:

Sprint Report	Date Range	Primary Focus	Key Deliverables & Milestones
WK1 Report	`2026-08-01` →\to `2026-08-07`	**Inception, LaTeX & Environment**	Custom `at_fuse_aif.cls` document class, Overleaf optimization, VS Code recipes, `README.md` baseline, initial `sync.ps1`.
WK2 Report	`2026-08-08` →\to `2026-08-14`	**M1 Schema Lock & Literature Matrix**	Milestone M1 schema contract (`schema.py` , `schema-lock-m1.md`), n<30n < 30 sample guards, 9-paper literature matrix in Walden format.
WK3 Report	`2026-08-15` →\to `2026-08-21`	**Requirements, Architecture & Standards**	Functional requirements (`FR-001` – `FR-005`), statutory mappings (EU AI Act / NIST AI RMF), 5-module system architecture, TA/Khwopa standards (Ruff, pyproject.toml), agent guidelines.
WK4 Report	`2026-08-22` →\to `2026-08-27` (Thursday close)	**Research Sprint, Core Engine & CLI**	20-track research sprint, 3-level synthesis, `CLAIM_LEDGER.md` , FairFace disk verification (97,698 images), Fairlearn + native AIF360 backends, BCa bootstrap (B≥1,000B \ge 1,000), χ2\chi^2 significance tests, SHAP explainer, `bias-aperture` CLI, 55/55 Pytest tests, multi-remote sync.

The current report after the August 27 cutoff is WK5 Report, covering August 28 through September 2 and the Phase-2 Product Upgrade Sprint.

---

3. High-Level Monthly Achievements

1. **Academic & Statutory Scaffolding**:
   - Proposal compiled and formatted using `at_fuse_aif.cls` (`report/main.pdf`).
   - Mapped all 5 functional requirements to EU AI Act (Articles 10, 13, 15) and NIST AI Risk Management Framework (Map, Measure, Manage).
2. **Empirical Groundwork & Claim Ledger**:
   - Resolved dataset disk counts: **97,698 released FairFace images** verified on disk.
   - Identified UTKFace synthetic DEX noise to support diagnostic boundaries.
   - Established auditable `CLAIM_LEDGER.md` tracking 60+ verified claims with formal reproduction procedures.
3. **Core Algorithmic Engine**:
   - Implemented Core Four metrics (DPD, DIR, EOP, EOD) with dual backends (Fairlearn and native AIF360 `BinaryLabelDataset` / `ClassificationMetric`).
   - Implemented 95% BCa Bootstrap Confidence Intervals ($B \ge 1,000$) and Chi-squared significance tests ($p < 0.05$).
   - Implemented exact linear Shapley surrogate feature attribution layer ($\phi_i = w_i(x_i - \mathbb{E}[x_i])$) in `explainability.py`.
4. **Tooling & Orchestration**:
   - Offline, single-file Jinja2 HTML report generator with embedded Base64 charts and zero CDN network calls.
   - Command-line interface (`bias-aperture audit`) with multi-backend and explainability toggles.
   - 100% pass rate across 55 automated unit tests (`uv run --extra dev pytest`).
   - Clean Git synchronization across all three configured remotes (`origin`, `duo`, `org`).

---

4. Master TO DOs & Remaining Roadmap

Priority	Category / Phase	Target Deliverable	Owner	Target Branch / File
☐ **P1**	**Empirical Benchmark**	Validation split inference (`predict.py`) over 10,954 images	Aaradhya	`data/processed/fairface_predicti`
☐ **P1**	**Empirical Benchmark**	Execute `bias-aperture audit` to render flagship HTML report	Aaradhya	`report/audit_report.html`
☐ **P1**	**Supervisor Logistics**	Confirm Viva defense date, duration, and score breakdown with TA Shreejan Kisee	Tisha	`docs/BiasAperture-AT.md`
☐ **P2**	**Compliance Verification**	Audit Model Card & Datasheet text for statutory compliance	Tisha	`report/audit_report.html`
☐ **P2**	**Data Validation**	Verify representation across 126 intersectional bins (7×9×27 \times 9 \times 2)	Tisha	`data/`
☐ **P2**	**LaTeX Report**	Ingest final benchmark tables and SHAP plots into `report/main.pdf`	Aaradhya	`report/main.pdf`
☐ **P3**	**Claim Ledger**	Transition ledger claims to `REPRODUCIBLE (LIVE BENCHMARK)`	Aaradhya	`docs/research/CLAIM_LEDGER.md`
☐ **P3**	**Defense Preparation**	Compile unified 10–15 min presentation deck & rehearse live demo	Joint	`docs/presentation/slides.pdf`

5. Related Master Documentation

- Master Task Division & Trait Allocation
- Session Walkthrough: Full Codebase Audit & Sync
- Task Division & Roadmap Audit

- Auditable Research Claim Ledger
- Proposal Defense Preparation Guide